BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 294 · Search engine indexing and PageRank

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a hyperlink?

Q2. What is an index in retrieval?

Q3. What kind of graph models links?

Q4. BST root20, left10, right30; insert 25. Where is it placed?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

crawler — Discovers/fetches documents.

inverted index — Maps terms to documents.

PageRank — Iterative link-based importance model.

damping — Probability factor in random-surfer model.

out-degree — Number of directed outgoing links.

New learning

Understand — A crawler discovers pages by following links and sitemaps; an index records useful information about documents for searching.

Apply — PageRank models a random web surfer and passes importance through directed links. A link from a high-ranked page can carry more weight than one from a low-ranked page.

Evaluate — PageRank uses damping and repeated iterations; raw incoming link counts alone are not the full algorithm. Search engines combine many ranking signals.

Worked example

In graph A→B and A→C, how is A's rank contribution divided (ignoring damping)?

1. Predict first — Outgoing degree of A is two, so its rank is divided across two outgoing edges.

2. Explain — Both B and C receive half of A's contribution before damping and other link contributions.

We Do · Practise with guidance

A has three outgoing links to B,C,D and rank 0.30. What raw contribution reaches C?

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0.30 / 3 = 0.10
Before damping/teleportation.
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Predict first

The outgoing links split A's contribution evenly in the simplified model.

Explain

C receives 0.10; final PageRank needs damping, all incoming contributions and repeated iterations.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Distinguish crawling and indexing. [4]

Question 2

In directed link graph A→B,C and B→C, identify incoming links of C and A's share given rank 0.4, ignoring damping. [6]

Question 3

Explain why the number of inbound links alone is not an adequate PageRank calculation. [6]

Exit Ticket · Answer before checking

A page's PageRank is simply its number of incoming links.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A reference to another page/resource.

Q2. A structure associating terms or keys with documents/records.

Q3. A directed graph.

Q4. 25>20 go right to30; 25<30 go left of30.

You Do pages · Independent questions

Q1. Crawling fetches/discovers pages [2]; indexing structures information for efficient query retrieval [2].

Q2. C receives from A and B [2]; A has two outgoing links [2]; A contributes 0.4/2=0.2 to C [2].

Q3. Importance weighted by linking pages [2]; outgoing degree splits contributions [1]; damping/teleportation [1]; repeated iterative convergence [1]; other ranking signals used in real search [1].

Exit Ticket · Model answer

False. Link weights, outgoing degree and damping matter.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.